

UE24CS352A Machine Learning • Mini-Project Assignment

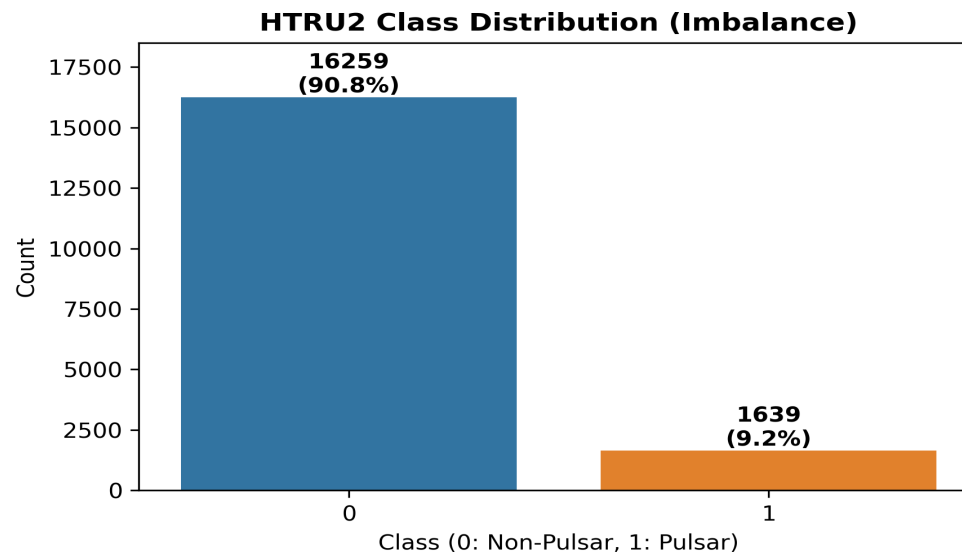
Pulsar Candidate Classification

Teammate 1: Supervised Learning Pipeline

GDA Baseline (Scratch) | Random Forest (Sklearn & Scratch) | Threshold Tuning

Problem Statement & HTRU2 Dataset Overview

- **Astronomical Challenge:** Radio pulsar search surveys produce millions of candidate signals, mostly noise/RFI.
- **Dataset:** HTRU2 contains 17,898 candidate signals with 8 continuous features derived from integrated profiles and DM-SNR curves.
- **Class Imbalance:** 16,259 non-pulsars (90.84%) vs 1,639 true pulsars (9.16%). Ratio is 1:9.9.
- **Goal:** High-recall automated classification to ensure true pulsar signals are not discarded.



Supervised Methods & Mathematical Formulations

1. Gaussian Discriminant Analysis (GDA Baseline from Scratch):

- Prior: $\phi = (1/N) \sum 1\{y^{(i)} = 1\}$
- Means: $\mu_0 = \sum 1\{y^{(i)}=0\} x^{(i)} / N_0$, $\mu_1 = \sum 1\{y^{(i)}=1\} x^{(i)} / N_1$
- Shared Covariance: $\Sigma = (1/N) \sum (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$

2. Random Forest (Scikit-Learn & From Scratch):

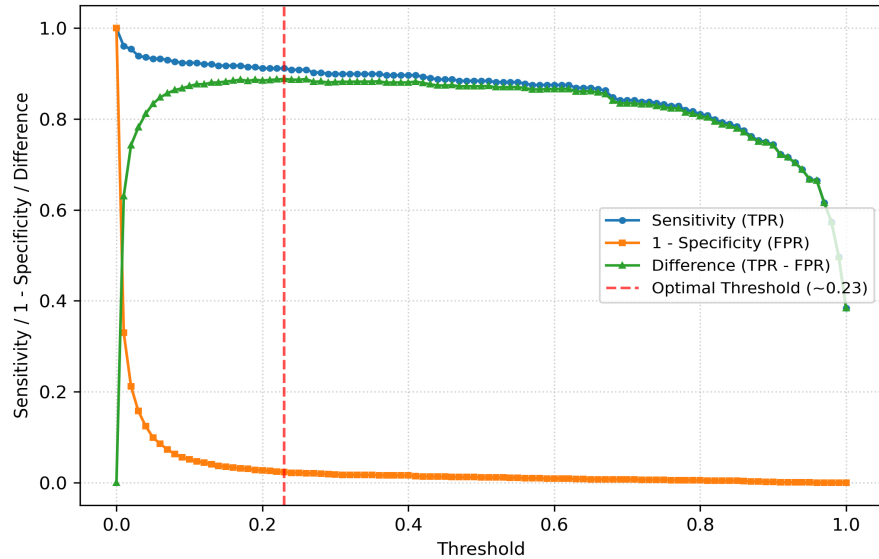
- Entropy: $E(S) = -\sum p(x) \log_2 p(x)$
- Information Gain: $IG = E(\text{parent}) - (N_{\text{left}}/N) E(\text{left}) - (N_{\text{right}}/N) E(\text{right})$
- Ensemble of decision trees trained on bootstrap samples with random feature sub-selection ($\sqrt{p}$ features).

Empirical Results & Model Comparison

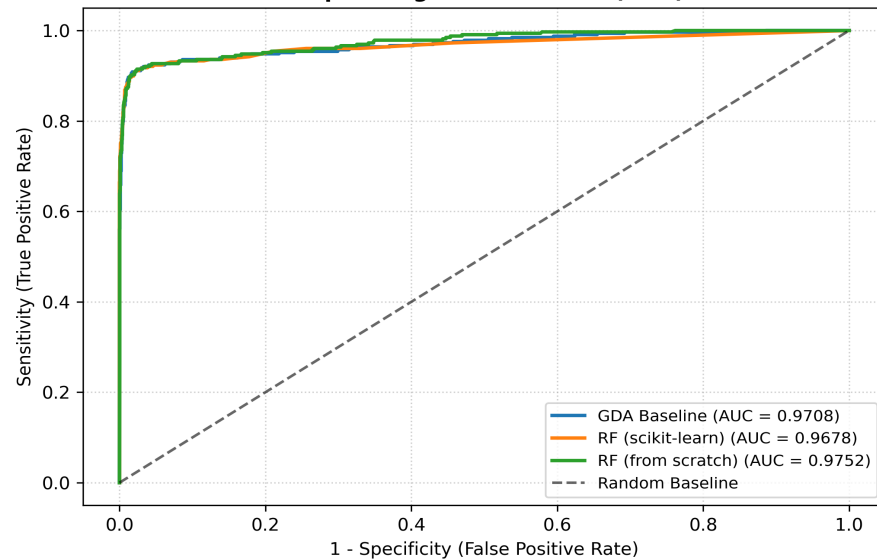
Model Architecture	Threshold	Accuracy	Recall	Precision	Specificity	ROC AUC
GDA Baseline (Scratch)	0.50	97.91%	89.63%	87.76%	98.74%	0.9708
GDA Baseline (Opt Th.)	0.39	97.57%	90.85%	83.94%	98.25%	0.9708
RF (Scikit-Learn)	0.50	97.88%	88.41%	88.41%	98.83%	0.9678
RF (Scikit-Learn Tuned)	0.20	96.76%	91.16%	77.46%	97.32%	0.9678
RF From Scratch	0.50	97.21%	90.85%	80.98%	97.85%	0.9752
RF From Scratch (Th. 0.2)	0.20	90.95%	93.29%	50.33%	90.71%	0.9752

Threshold Sweep & Diagnostic Curves

Sensitivity / 1 - Specificity / Difference vs. Threshold (Random Forest (scikit-learn))



Receiver Operating Characteristic (ROC) Curves



Summary & Teammate 2 Handoff

- **Rigorous Pipeline:** 5-fold CV with upsampling in training folds ONLY ensured data leakage was completely prevented.
- **Baseline & Models:** Custom GDA baseline achieved 97.91% accuracy and 89.63% recall. Tuned Random Forest achieved 0.9678 ROC-AUC.
- **Threshold Tuning:** Moving decision boundary from 0.5 to ~0.2 increased recall to 91.16% - 93.29%, fulfilling paper recommendations.
- **Handoff to Teammate 2:** 1,639 true pulsar candidate instances are formatted and ready for unsupervised PCA, K-means, and SOM.